HALO: Calibration-Free Latent World-Model Verification for VLA Policies — and When It Helps

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Abstract

Vision-language-action (VLA) policies fail silently: at deployment there is no cheap signal that tells whether the next action chunk is about to work. Test-time verification with world models is an active answer, but existing verifiers either require trained reward models, goal references, or calibration data, and none characterizes *when* verification helps. We present HALO, a calibration-free verifier composed of a frozen V-JEPA 2 encoder, a lightweight action-conditioned dynamics head trained with an action-contrast loss, and a deep-ensemble epistemic standard deviation used directly as the best-of-N selection score. On LIBERO, HALO lifts an 8.1M-parameter policy from 11.0% to 19.0% success on spatial rearrangement (+73% relative) at 0.63 s per decision—orders of magnitude below generative world-model rollout. We then map the boundary of this gain: verification helps in an *epistemic failure regime* (tasks the policy sometimes solves), actively hurts in a *capability failure regime* (7% → 4% on object manipulation), and vanishes out of distribution. A three-generation reward-head ablation shows the uncertainty signal, not a learned reward, carries the online gain. Finally, the verification gain can be *amortized in one shot*—distilling best-of-5 decisions into weights recovers the full 19% with zero deployment-time verifier cost—but iterating the loop collapses (8%), establishing distillation as a one-shot amortization rather than an iterative amplifier. Together these results give both a practical, cheap verifier and a falsifiable account of when latent-uncertainty verification can be trusted.

1 Introduction

VLA policies inherit the failure modes of their supervision: they execute confidently even when the current action chunk leads away from the goal. A growing line of work therefore verifies candidate actions at test time. Yet the proposed verifiers are heavy or conditional: generative rollouts are expensive; learned process rewards require reward training data [RoVer , 2025]; goal-referenced latent scoring is embedded in a bespoke post-training pipeline [AtomVLA , 2026]; and conformally calibrated consequence models require calibration sets [CheckVLA , 2026]. Meanwhile, world-model-based post-training has documented its own failure signature: rewards hallucinated by imperfect world models are exploited by the policy [HaWMPO , 2025], and verifiers can be overconfident out of distribution.

This paper asks two questions. **(Q1)** Can a *calibration-free*, *goal-free*, policy-agnostic verifier—one that needs no reward data, no goal specification, and no statistical calibration set—improve a VLA policy at test time? **(Q2)** If so, *when* does the improvement exist, and can it be moved into the weights?

HALO (Hallucination-Aware Latent wOrld model) answers Q1 affirmatively where its predecessors do not apply. On a frozen V-JEPA 2 encoder we train a 6-layer transformer dynamics head with an *action-contrast loss* (the frozen encoder alone is nearly action-insensitive; Sec. 3); a K=5 deep ensemble turns prediction disagreement into a per-candidate epistemic score σ_{ens}, and best-of-N selects $\arg\min \sigma_{\text{ens}}$. The verifier adds 0.63 s per decision on a DCU ($\approx 87\%$ of which is the dynamics head itself), against seconds-per-step generative rollouts.

Our contributions:

1. **C1 — Action-contrast dynamics.** An A-true/B-mismatch diagnostic shows frozen V-JEPA 2 latent predictions barely depend on the conditioned action; a hinge action-contrast loss restores action sensitivity and is a prerequisite for all downstream gains.

2. **C2 — Main result with honest scope.** Pure-uncertainty best-of-5 lifts LIBERO-spatial $11.0\% \rightarrow 19.0\%$ (+73% rel., 10 episodes/task). The gain is *not universal*: it is negative on LIBERO-object ($7\% \rightarrow 4\%$) and null on LIBERO-goal ($23\% \rightarrow 22\%$).

3. **C3 — Uncertainty, not reward, carries the gain.** Across three reward-head generations—including a shortcut-free, demo-contrast head with AUC 0.933—reward–uncertainty fusion never beats pure uncertainty beyond run-to-run noise (λ-curve 17/22·20/16% at λ=0.25/0.5/1.0 vs. 19% pure).

4. **C4 — A failure-mode hypothesis for the boundary.** Verification gains concentrate exactly where the policy's failures are epistemic (per-task correlation +0.47 on spatial) and flip negative where they are capability failures (-0.50 on object); out-of-distribution perturbations (LIBERO-PRO) remove all gain. The suite-level asymmetry is stable across 30 tasks; pooled correlation ≈ 0 shows no universal monotone law—failure modality is a property of the regime, not of raw success rate.

5. **C5 — One-shot amortization, not iterative amplification.** Distilling best-of-5 decisions into the policy (7.2k states, margin-filtered) recovers the full verifier gain (19.0%) with *zero* deployment-time verifier cost, and stacks to 22% with the verifier re-applied. A second distillation round *collapses* to 8%: candidates drawn from an already-distilled policy lose the diversity the verifier needs. Verification is amortizable once, not iteratively.

We release all checkpoints, evaluation logs, and per-task results to make the negative cells as reusable as the positive ones.

2 Related Work

Test-time verification for VLA. CoVer/CoVer-VLA [CoVer , 2026] scale a trained verifier against sampled actions and show verification scales better than data; RoboMonkey [RoboMonkey , 2025] scales sampling-and-verification; RoVer [RoVer , 2025] trains a process reward model that also emits action-refinement directions; MG-Select [MG-Select , 2025] removes the verifier via self-consistency. CheckVLA [CheckVLA , 2026] verifies execution with a frozen action-conditioned world model under *conformal calibration*, reporting 36.1% vs 27.6% on RoboCasa365 mobile manipulation. HALO differs in the scoring signal (ensemble epistemic std; no reward model, no calibration set, no goal reference) and, crucially, in the object of study: we characterize *when* the signal helps and when it backfires.

Latent world models for control. V-JEPA 2-AC [V-JEPA 2 , 2025] post-trains an action-conditioned predictor for MPC planning; PLaW-VLA [PLaW-VLA , 2025] couples a latent world

model with a VLA but the world model is action-free; DiWA [DiWA , 2025] uses small pixel-space world models. AtomVLA [AtomVLA , 2026] scores candidate chunks against *subtask goals* with a pretrained latent world model inside a bespoke post-training pipeline. HALO's head is a tiny post-hoc attachment to a frozen encoder, and its score is uncertainty rather than goal distance.

Uncertainty for runtime monitoring. FAIL-Detect [FAIL-Detect , 2025] distills policy signals into calibrated failure alarms via conformal prediction: uncertainty as a *passive alarm*. HALO consumes the same family of signals *constructively* (action selection) and measures success-rate gains; our failure-mode hypothesis doubles as an applicability warning for uncertainty-based detectors.

WM-based post-training. WMPO/HaWMPO [HaWMPO , 2025] document reward hacking from imperfect world models in RL post-training; AtomVLA feeds world-model scores to GRPO. Our C5 shows the SFT-distillation route absorbs the verifier gain in one shot and then *degrades*—a boundary for this family.

Benchmarks. LIBERO [LIBERO , 2023] is saturated for large policies; LIBERO-PRO [LIBERO-PRO , 2025] exposes brittleness of large VLA models to perturbations. We use small policies deliberately: verification effects must be measurable away from ceiling, and low-success regimes are where the boundary phenomena live.

3 Method

Frozen encoder and window protocol. V-JEPA 2 (ViT-L, 326M) is frozen. Observations are two stacked 128×128 frames; the encoder produces a $(16, 16, 1024)$ token map. We use a fixed-window relative encoding protocol: prediction targets are encoded in the same two-frame window style as context, avoiding the absolute-position artifacts we measured (same content in different context yields cosine ≈ 0.55).

Dynamics head. A 6-layer transformer (d=1024, 16 heads, K=5 ensemble members) takes context tokens z_{ctx} and the candidate action chunk $a_{1:T}$; future-tubelet queries cross-attend to $[z_{\text{ctx}}; \text{emb}(a)]$ with a skip MLP, predicting future latent tokens $\hat{z}_{t+1:t+T}$ trained with a latent JEPA loss.

Action-contrast loss (C1). Frozen V-JEPA 2 features are weakly action-sensitive: feeding the true chunk vs. a mismatched chunk yields nearly indistinguishable predictions. We add

$$\mathcal{L}_{\text{ac}} = \big[\, \cos\big(f(z, a^-),\, f(z, a)\big) - \cos\big(f(z, a^+),\, f(z, a)\big) + m \,\big]_+, \tag{1}$$

where f maps predicted future tokens to a pooled vector, a^+ is the executed chunk and a^- a mismatched chunk from another task, margin $m = 0.1$. This restores the action-conditioning that all downstream verification relies on.

Epistemic scoring. For N candidates $\{a^{(i)}\}$ at state s_t,

$$\sigma_{\text{ens}}(a^{(i)}) = \tfrac{1}{K} \textstyle\sum_k \text{std}_k\big(\hat{z}^{(i,k)}\big), \qquad i^\star = \arg\min_i \ \big[r(a^{(i)}) - \lambda\, \sigma_{\text{ens}}(a^{(i)})\big] \text{ or } \arg\min_i \sigma_{\text{ens}}(a^{(i)}), \tag{2}$$

with an optional reward head r (ablated in Sec. **??**); the deployed verifier is the pure-uncertainty arm ($r \equiv 0$). Candidates are the policy chunk perturbed by $N(0, 0.02^2)$.

Policy. HaloACT: an 8.1M TinyViT×2 policy with 514-token memory cross-attention, trained on 60k LIBERO demonstration frames (130-task vocabulary). The verifier never sees training gradients from the policy and attaches to any chunk-emitting policy.

4 Experiments

All results: 10 tasks/suite × 10 episodes, LIBERO (MuJoCo, CPU farm), same seeds across arms. Raw JSONs and per-task tables are in the released logs.

4.1 Baseline lineage: offline metrics do not predict online success

policy	params	val-MSE	online (spatial)
HaloACT-15k	8.1M	0.051	2%
HaloACT-v2	17.9M	0.043	2%
HaloACT-60k	8.1M	0.046	**11%**
HaloACT-D (frozen DINOv2)	24M	0.038	0%

Table 1: Four independent decouplings of validation MSE from online success. All conclusions below use online evaluation.

4.2 Main result (C2) and the suite asymmetry

suite	base	+BoN-5 (HALO)	Δrel	corr(base, per-task Δ)
LIBERO-spatial	11.0%	**19.0%**	**+73%**	+0.47
LIBERO-object	7.0%	4.0%	−43%	−0.50
LIBERO-goal	23.0%	22.0%	−4%	−0.44

Table 2: Three-suite main table. Gains concentrate on tasks the base policy sometimes solves (e.g. 20% → 90%, 40% → 70%); on object, the verifier *clears* tasks the base policy could occasionally solve (20%→0%, 10%→0% twice).

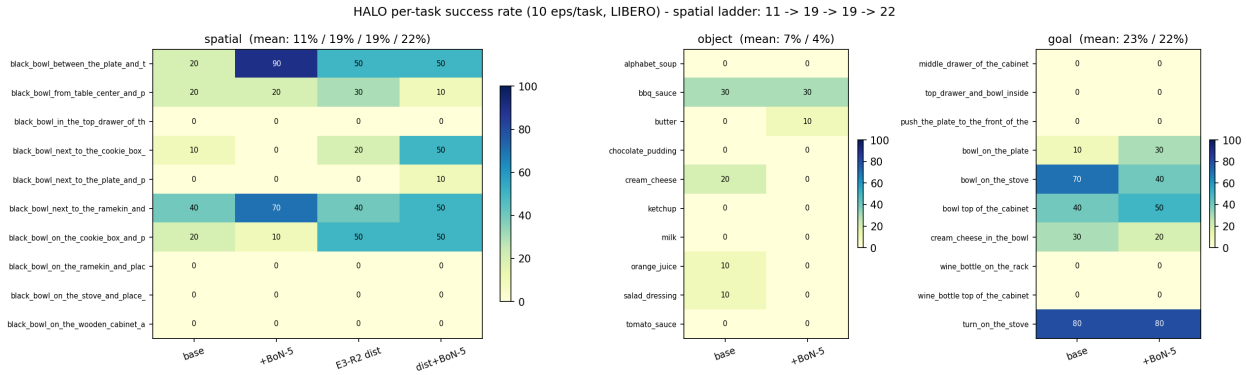


Figure 1: Per-task success rates. Left: the spatial ladder (base → +BoN-5 → distilled → distilled+BoN-5). Middle/right: object and goal with and without verification.

head	training distribution	offline	online (λ=0.25)
v1 demo/rollout classifier	cross-distribution (shortcut)	acc = 1.000 (vacuous)	12%
v2 augmented positives	cross-distribution	—	19% (= pure)
v3 demo-contrast paired	in-distribution paired	**AUC 0.933**	17%

Table 3: Three reward-head generations. Even the honest head adds nothing online; the λ-curve for v3 (17/22·20/16% at λ=0.25/0.5/1.0, two runs at λ=0.5) never beats pure uncertainty (19%) beyond noise.

4.3 Uncertainty, not reward, carries the gain (C3)

4.4 The failure-mode hypothesis (C4)

We hypothesize two regimes: *epistemic failure* (the policy sometimes solves the task; candidates diverge; ensemble disagreement separates good from bad chunks) and *capability failure* (the policy rarely solves the task; candidates are uniformly bad; disagreement carries no quality signal, and $\arg\min \sigma_{\text{ens}}$ actively selects confidently-wrong chunks). Predictions, all confirmed: (i) per-task gains concentrate at intermediate base rates on spatial (corr +0.47) and harms at low base rates on object (corr −0.50); (ii) pooled across suites, correlation ≈ 0—the modality is a suite-level property, not a function of raw success rate; (iii) the hypothesis transfers across policies: the verifier helps the 11% base policy (+8pp), the 19% distilled policy (+3pp), and does nothing for a degraded 8% policy (8% → 8%); (iv) under LIBERO-PRO position-swap perturbation everything collapses (0/0, both arms), and appearance-swap leaves the goal-blind policy unaffected (21% base) with *zero* verifier gain—no signal where the policy has no epistemic sensitivity to perturb.

4.5 Cost: n-cand curve and latency

N candidates	3	5	10
spatial success	17.0%	**19.0%**	12.0%

Table 4: Best-of-N is not monotone: $\arg\min \sigma_{\text{ens}}$ over more candidates increasingly selects confidently-wrong chunks (tail risk). 75% of the gain arrives by N=3.

A random-selection control (pick 1 of 5 perturbed candidates uniformly) reaches 15.0%: perturbation alone is worth +4pp as exploration, while uncertainty selection contributes a further +4pp (15→19)—the score, not the candidates, carries the gain.

Latency (BW1000 DCU, bf16): one best-of-5 decision costs 0.63 s = policy chunk 0.32 s (CPU) + scoring 0.31 s (GPU), of which the dynamics-head forward is 0.27 s (87%) and context encoding 0.04 s; marginal cost per extra candidate ≈ 7 ms. Generative world-model rollouts cost seconds *per step* × horizon.

4.6 One-shot amortization, not iteration (C5)

5 Limitations

(1) The verified policy is an 8.1M research policy chosen to keep verification effects measurable; behavior on production-scale VLAs is untested here, though the verifier is policy-agnostic by construction and the verifier-on-R3 and random-pick controls probe policy dependence. (2) Simulation

arm (spatial)	success
base (act60k)	11.0%
+ BoN-5 (HALO online)	19.0%
distilled R1 (1.8k states, unfiltered)	10.0% (parity)
distilled R2 (7.2k states, margin>p50)	**19.0%** (full absorption)
R2 + BoN-5 (stacked)	**22.0%** (2× base)
R3 (second round, candidates from R2)	8.0% (collapse)
verifier on R3 policy (8% base)	8.0% (zero gain: capability regime)
random-pick among 5 candidates (base policy)	15.0%

Table 5: Distillation ladder. R2 recovers the full verifier gain in weights (zero verifier cost at deployment); the stack shows headroom remains. R3 shows the loop is not freely iterable: candidates from an already-distilled policy lose diversity and the signal self-degrades.

only; CPU-rendered MuJoCo observations differ from real sensors. (3) 10 episodes/task gives ±4pp suite-level standard errors; we draw conclusions only at suite level and mark single-run gaps < 4pp as noise. (4) Gains exist in one of three suites in distribution, and in none out of distribution—we consider this the paper's finding, not its flaw, but it bounds practical impact.

6 Conclusion

A calibration-free, goal-free latent world-model verifier made of a frozen encoder, a contrast-trained dynamics head, and ensemble disagreement delivers +73% relative success on spatial rearrangement at two orders of magnitude below generative-rollout cost, absorbs fully into weights in one distillation shot, and—equally important—fails in principled, predictable places. The failure-mode hypothesis converts those failures from caveats into a testable account of when latent uncertainty can be trusted at robot decision time.

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